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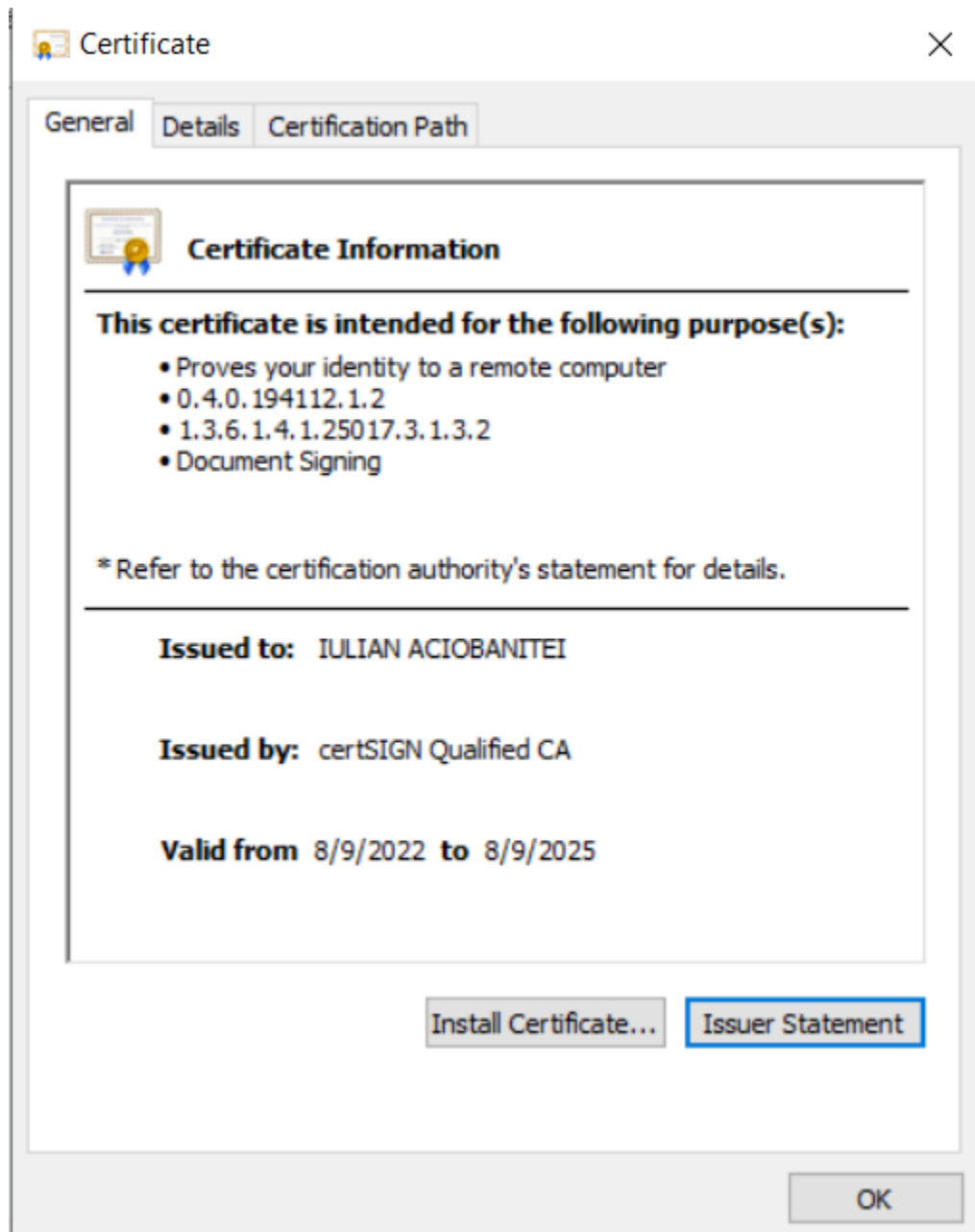
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Electronic Signatures - Tasks

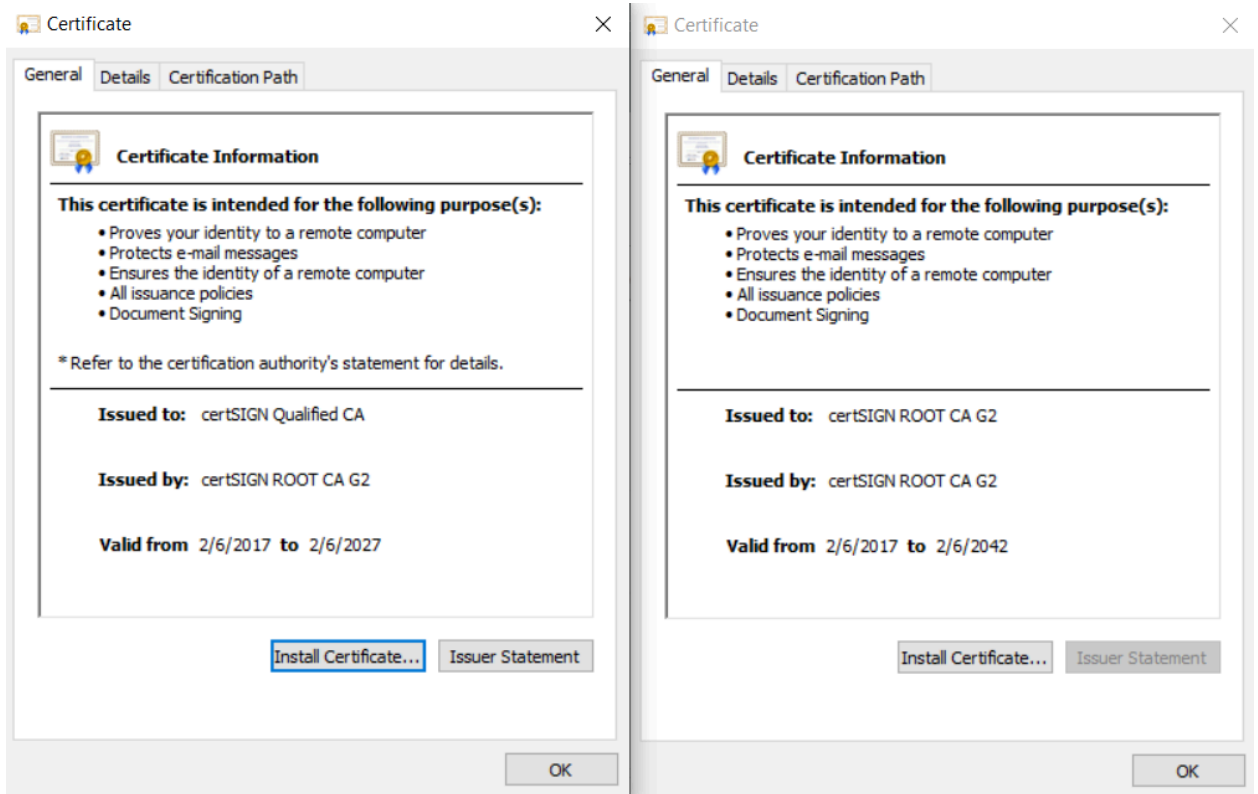
1. Certificates

1.1. Download this certificate

(<https://drive.google.com/file/d/16KXExKkzw4b35OedxrJ7hAA6YTLbQk72/view>)

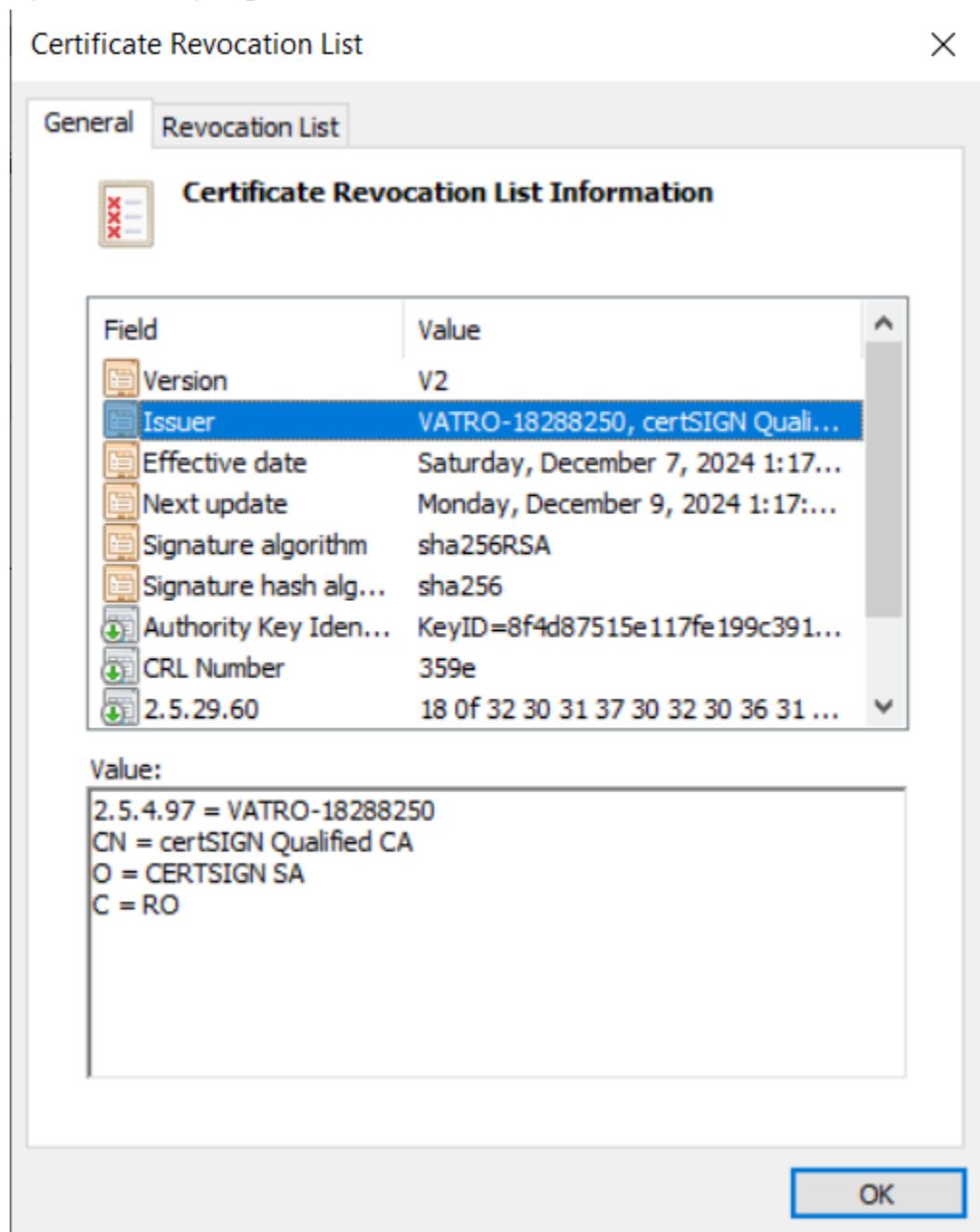


1.2. Download all the certificates in the certification path



1.3. Download the CRL manually

<http://crl.certsign.ro/certsign-qualifiedca.crl>



1.4. Verify the certificate using certutil and CRL

```
C:\Users\Angi\Downloads>certutil -URL cert.cer
```

URL Retrieval Tool

Status	Type	Url	Retrieval Time	Thumbprint
Verified	Base CRL (...)	[0.0] http://crl.certsign.ro/certsign-qu...	1	3199b3752...

Timeout (sec) Note: CRLs or certificates being downloaded are not exhaustively verified. A CRL or cert may still be inconsistent or may not have the proper extensions to allow for correct verification.

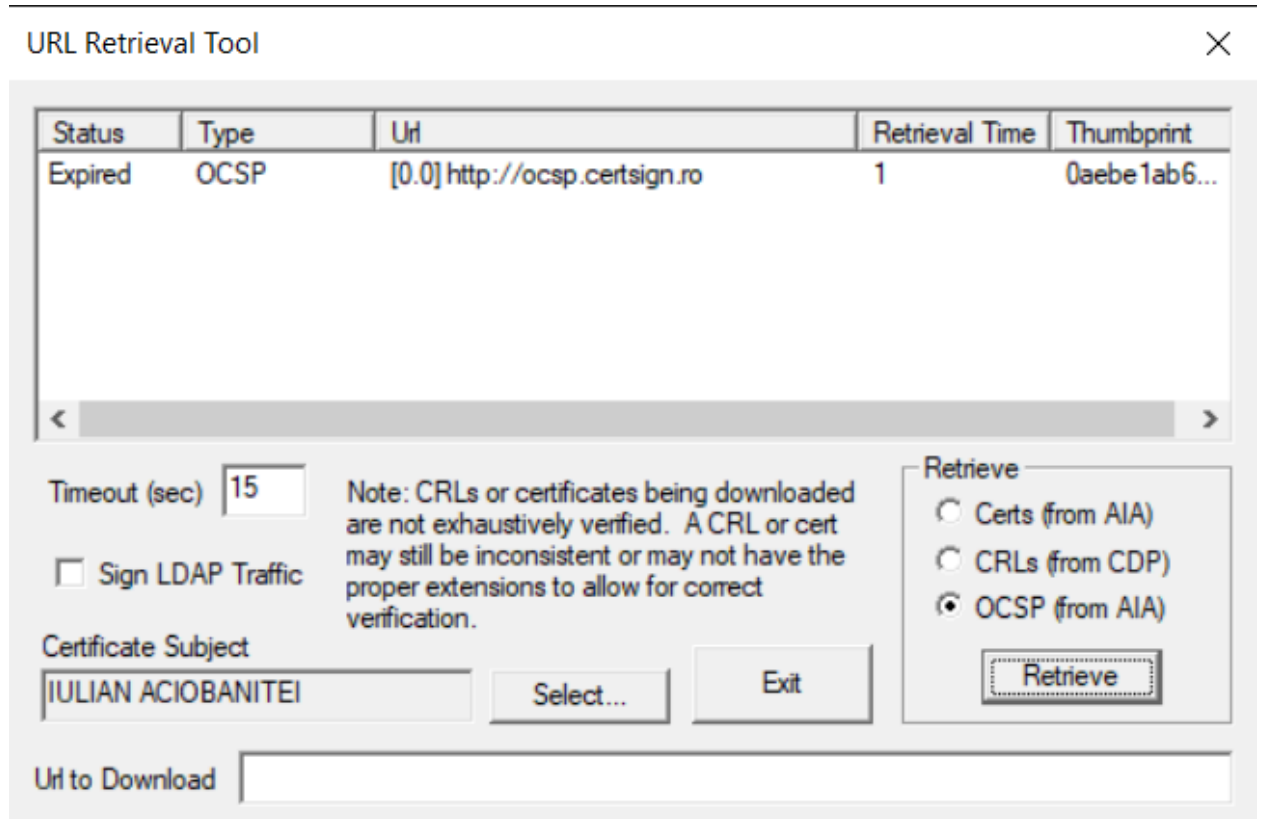
☐ Sign LDAP Traffic

Certificate Subject

Retrieve
☐ Certs (from AIA)
☒ CRLs (from CDP)
☐ OCSP (from AIA)

Url to Download

1.5. Verify the certificate using certutil and OCSP



1.6. Verify the certificate using openssl*

*Example of openssl cmd(admin): openssl ocspl -issuer ca_cert.cer -cert cert.cer -no_nonce -url "ocsp url" -CAfile root.cer

openssl ocspl -issuer certsign-qualifiedca.crt -cert cert.cer -no_nonce -url "http://ocsp.certsign.ro/" -CAfile certsign-rootg2.crt

```
C:\Users\Angi\Downloads>openssl ocspl -issuer certsign-qualifiedca.crt -cert cert.cer -no_nonce -url "http://ocsp.certsign.ro/" -CAfile certsign-rootg2.crt
Response verify OK
cert.cer: good
    This Update: Dec  7 19:13:18 2024 GMT
    Next Update: Dec  8 19:13:18 2024 GMT
```

1.7. View certificate using an ASN.1 viewer (openssl cmd or lapo.it)

```
C:\Users\Angi\Downloads>openssl asn1parse -in cert.cer -inform DER
 0:d=0  hl=4  l=1781 cons: SEQUENCE
 4:d=1  hl=4  l=1245 cons: SEQUENCE
 8:d=2  hl=2  l= 3 cons: cont [ 0 ]
10:d=3  hl=2  l= 1 prim: INTEGER           :02
13:d=2  hl=2  l=12 prim: INTEGER           :220936B818E89FF45D5C2664
27:d=2  hl=2  l=13 cons: SEQUENCE
29:d=3  hl=2  l= 9 prim: OBJECT            :sha256WithRSAEncryption
40:d=3  hl=2  l= 0 prim: NULL
42:d=2  hl=2  l=92 cons: SEQUENCE
44:d=3  hl=2  l=11 cons: SET
46:d=4  hl=2  l= 9 cons: SEQUENCE
48:d=5  hl=2  l= 3 prim: OBJECT            :countryName
53:d=5  hl=2  l= 2 prim: PRINTABLESTRING  :RO
57:d=3  hl=2  l=20 cons: SET
59:d=4  hl=2  l=18 cons: SEQUENCE
61:d=5  hl=2  l= 3 prim: OBJECT            :organizationName
66:d=5  hl=2  l=11 prim: PRINTABLESTRING  :CERTSIGN SA
79:d=3  hl=2  l=30 cons: SET
81:d=4  hl=2  l=28 cons: SEQUENCE
83:d=5  hl=2  l= 3 prim: OBJECT            :commonName
88:d=5  hl=2  l=21 prim: PRINTABLESTRING  :certSIGN Qualified CA
111:d=3  hl=2  l=23 cons: SET
113:d=4  hl=2  l=21 cons: SEQUENCE
115:d=5  hl=2  l= 3 prim: OBJECT            :organizationIdentifier
120:d=5  hl=2  l=14 prim: PRINTABLESTRING  :VATRO-18288250
136:d=2  hl=2  l=30 cons: SEQUENCE
138:d=3  hl=2  l=13 prim: UTCTIME          :220809093106Z
153:d=3  hl=2  l=13 prim: UTCTIME          :250809093106Z
168:d=2  hl=3  l=140 cons: SEQUENCE
171:d=3  hl=2  l=11 cons: SET
173:d=4  hl=2  l= 9 cons: SEQUENCE
175:d=5  hl=2  l= 3 prim: OBJECT            :countryName
180:d=5  hl=2  l= 2 prim: PRINTABLESTRING  :RO
184:d=3  hl=2  l=20 cons: SET
186:d=4  hl=2  l=18 cons: SEQUENCE
188:d=5  hl=2  l= 3 prim: OBJECT            :organizationName
193:d=5  hl=2  l=11 prim: UTF8STRING       :CERTSIGN SA
206:d=3  hl=2  l=27 cons: SET
208:d=4  hl=2  l=25 cons: SEQUENCE
210:d=5  hl=2  l= 3 prim: OBJECT            :commonName
215:d=5  hl=2  l=18 prim: UTF8STRING       :IULIAN ACIOBANITEI
235:d=3  hl=2  l=14 cons: SET
237:d=4  hl=2  l=12 cons: SEQUENCE
239:d=5  hl=2  l= 3 prim: OBJECT            :serialNumber
244:d=5  hl=2  l= 5 prim: PRINTABLESTRING  :AI916
251:d=3  hl=2  l=15 cons: SET
253:d=4  hl=2  l=13 cons: SEQUENCE
255:d=5  hl=2  l= 3 prim: OBJECT            :givenName
260:d=5  hl=2  l= 6 prim: UTF8STRING       :IULIAN
```


1.8. Extract the OID of an attribute from Subject extension, at will.

```
C:\Users\Angi\Downloads>openssl x509 -in cert.cer -noout -text | findstr "Issuer"
    Issuer: C=RO, O=CERTSIGN SA, CN=certSIGN Qualified CA, organizationIdentifier=VATRO-18288250
    CA Issuers - URI:http://www.certsign.ro/certcrl/certsign-qualifiedca.crt
```

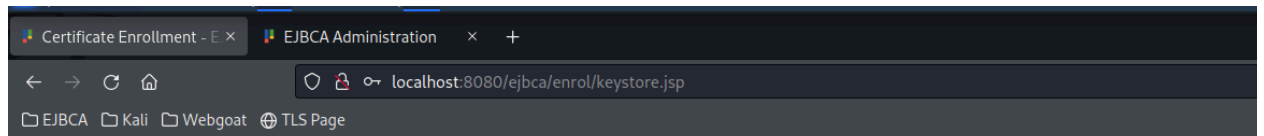
2. Get a signing certificate

2.1. From EJBCA's admin interface, create a new end entity of type User_Sign or TLS_Client

The screenshot shows the EJBCA Administration web interface in a browser. The address bar indicates the URL is `https://localhost:8443/ejbcadminweb/ra/addendentity.jsp`. The page title is "Add End Entity". On the left, there is a navigation menu with sections: "CA Functions" (including CA Activation, CA Structure & CRLs, Certificate Profiles, Certification Authorities, Crypto Tokens, Publishers, and Validators), "RA Functions" (including Add End Entity, End Entity Profiles, Search End Entities, and User Data Sources), "Supervision Functions" (including Approval Profiles and Approve Actions), "System Functions" (including Roles and Access Rules, Internal Key Bindings, and Services), "System Configuration" (including CMP Configuration, EST Configuration, SCEP Configuration, and System Configuration), and "My Preferences". The main content area is the "Add End Entity" form, which is divided into several sections: "End Entity Profile" (with a dropdown menu set to "User"), "Username" (text field with "angelica"), "Password (or Enrollment Code)" (password field with 6 dots), "Confirm Password" (password field with 6 dots), "E-mail address" (text field with an @ symbol), "Subject DN Attributes" (with fields for CN, Common name, givenName, Given name (first name), surname, Surname (last name), C, Country (ISO 3166), and O, Organization), and "Main Certificate Data" (with fields for Certificate Profile, CA, and Token). Each field has a "Required" checkbox on the right. The "Required" checkboxes are checked for "Username", "Password (or Enrollment Code)", "Confirm Password", "CN, Common name", "C, Country (ISO 3166)", "Certificate Profile", "CA", and "Token". The "E-mail address" checkbox is unchecked. The "Add" and "Reset" buttons are at the bottom of the form.

Add End Entity		Required
End Entity Profile	User	
Username	angelica	☒
Password (or Enrollment Code)		☒
Confirm Password		☒
E-mail address	@	☐
Subject DN Attributes		
CN, Common name	Angelica F.	☒
givenName, Given name (first name)	Angelica	☐
surname, Surname (last name)	F.	☐
C, Country (ISO 3166)	RO	☒
O, Organization	ASW	☐
Main Certificate Data		
Certificate Profile	User_Sign	☒
CA	IssuingCA	☒
Token	P12 file	☒
Add Reset		

2.2. From EJBCA's public interface, actually generate the certificate and obtain the .p12 file.



Enroll

Create Certificate from CSR
Create Keystore
Create CV certificate

Register

Request Registration

Retrieve

Fetch CA Certificates
Fetch CA CRLs
List User's Certificates
Fetch User's Latest Certificate

Inspect

Inspect certificate/CSR
Check Certificate Status

Miscellaneous

Administration

The EJBCA Public Web has been deprecated and will be removed in an upcoming version of EJBCA. Please move your workflow to the new EJBCA Public Web.

Keystore Enrollment

Welcome to Keystore Enrollment.

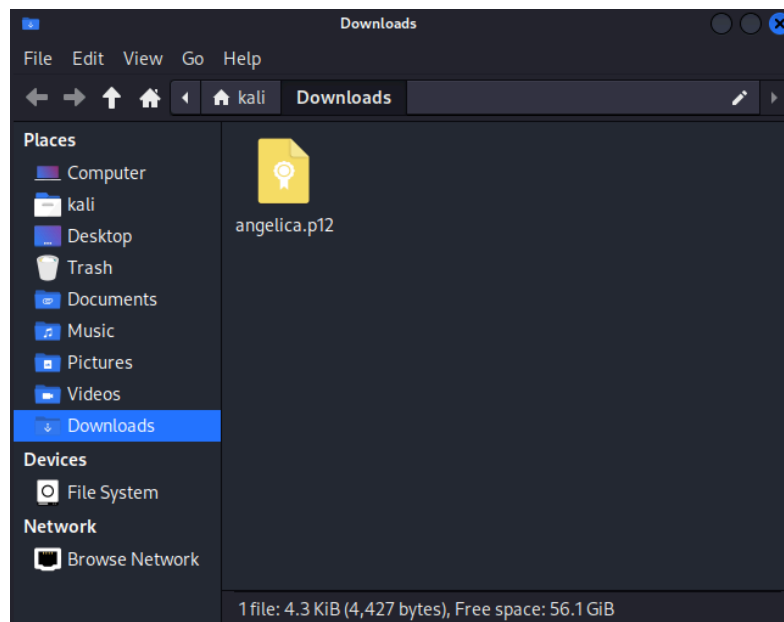
Please enter your username and password. Then click OK to generate your token.

Authentication

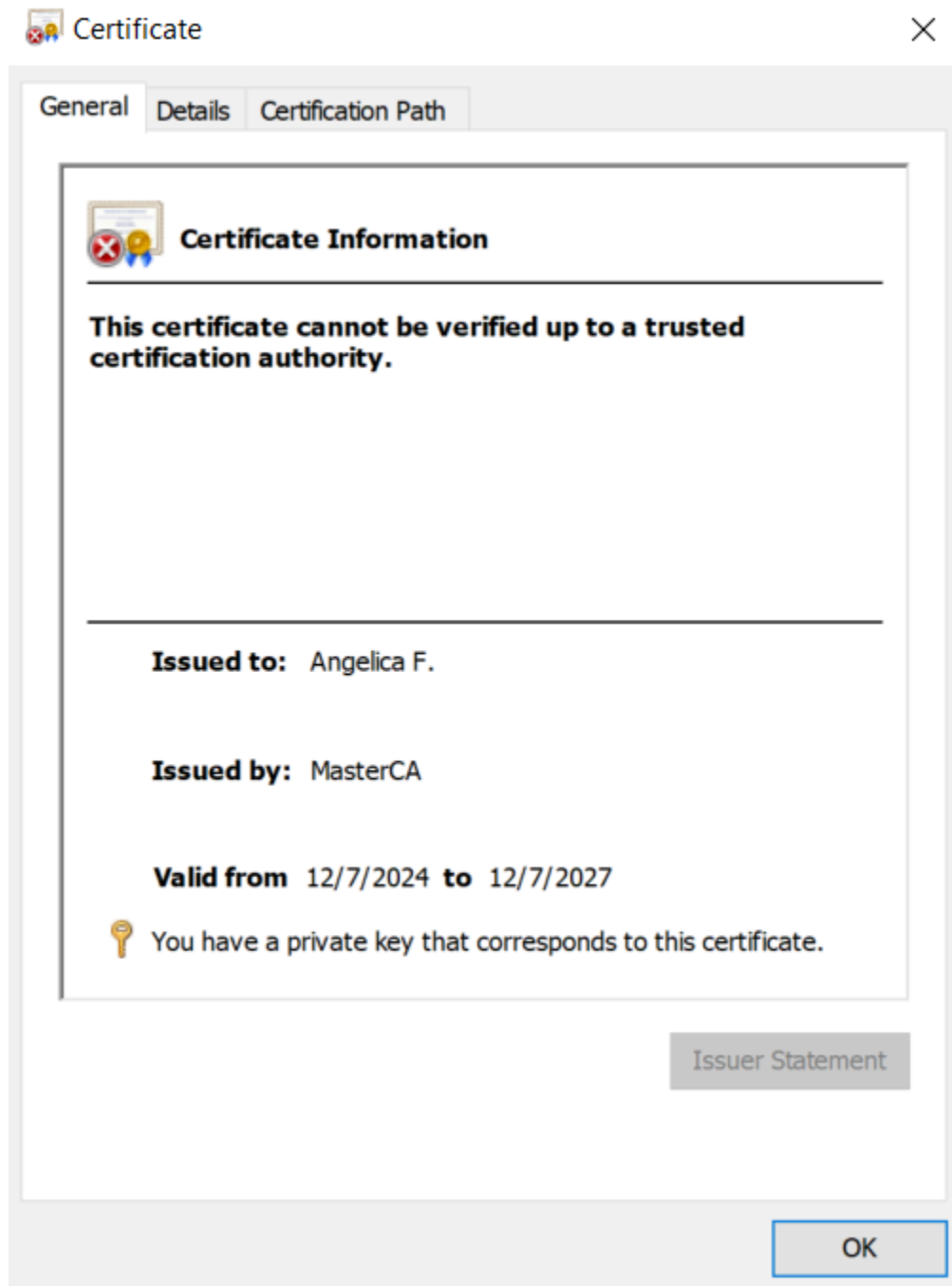
Username angelica

Password •••••

OK



2.3. Import the generated certificate into the Windows Store.



3. PDF Signatures

3.1. Sign a document

Semnați cu un ID digital

Alegeți ID-ul digital pe care doriți să îl utilizați pentru semnare:

Angelica F. (ID digital Windows)
Emis de: MasterCA, Expiră: 2027.12.07

Reîmprospătare

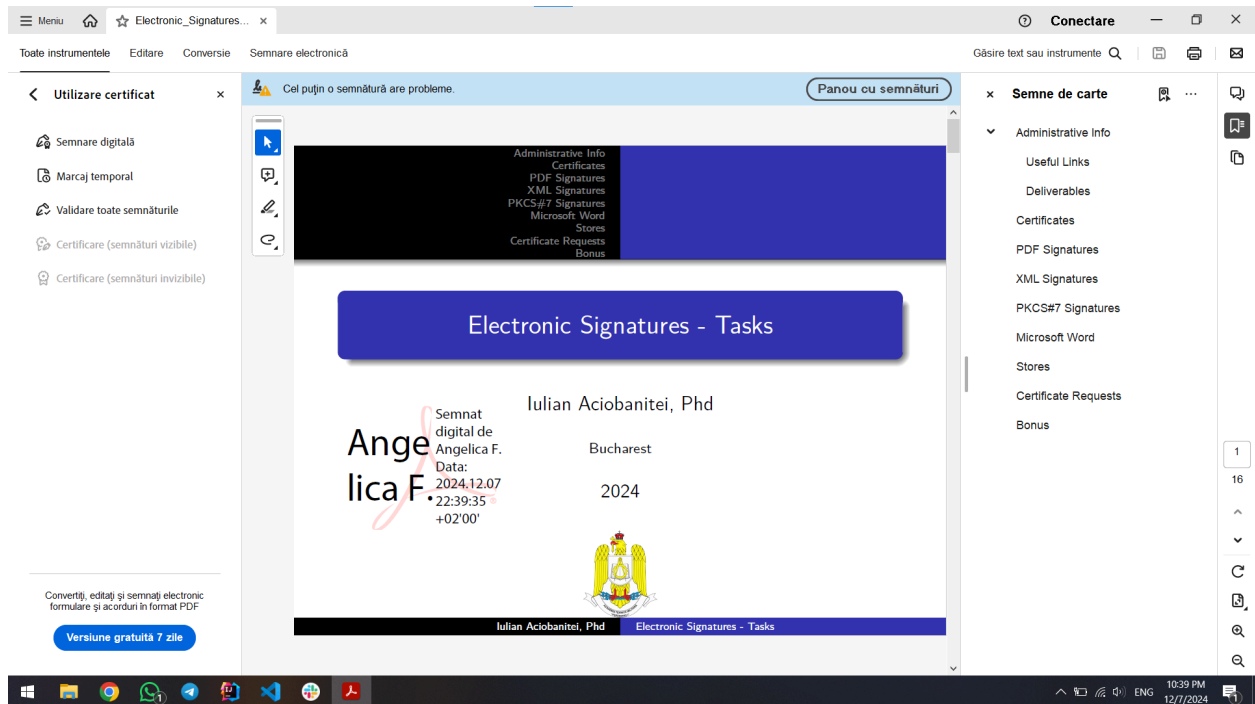
Vizualizare detalii

?

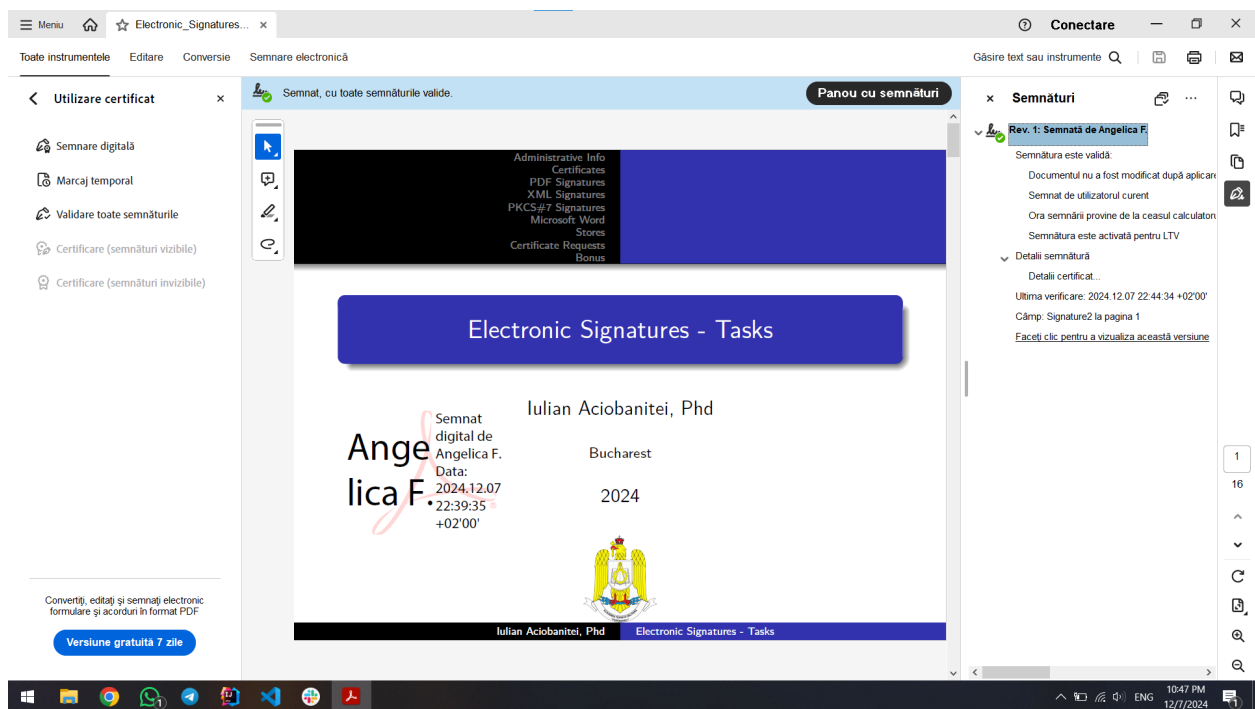
Configurare ID digital nou

Anulare

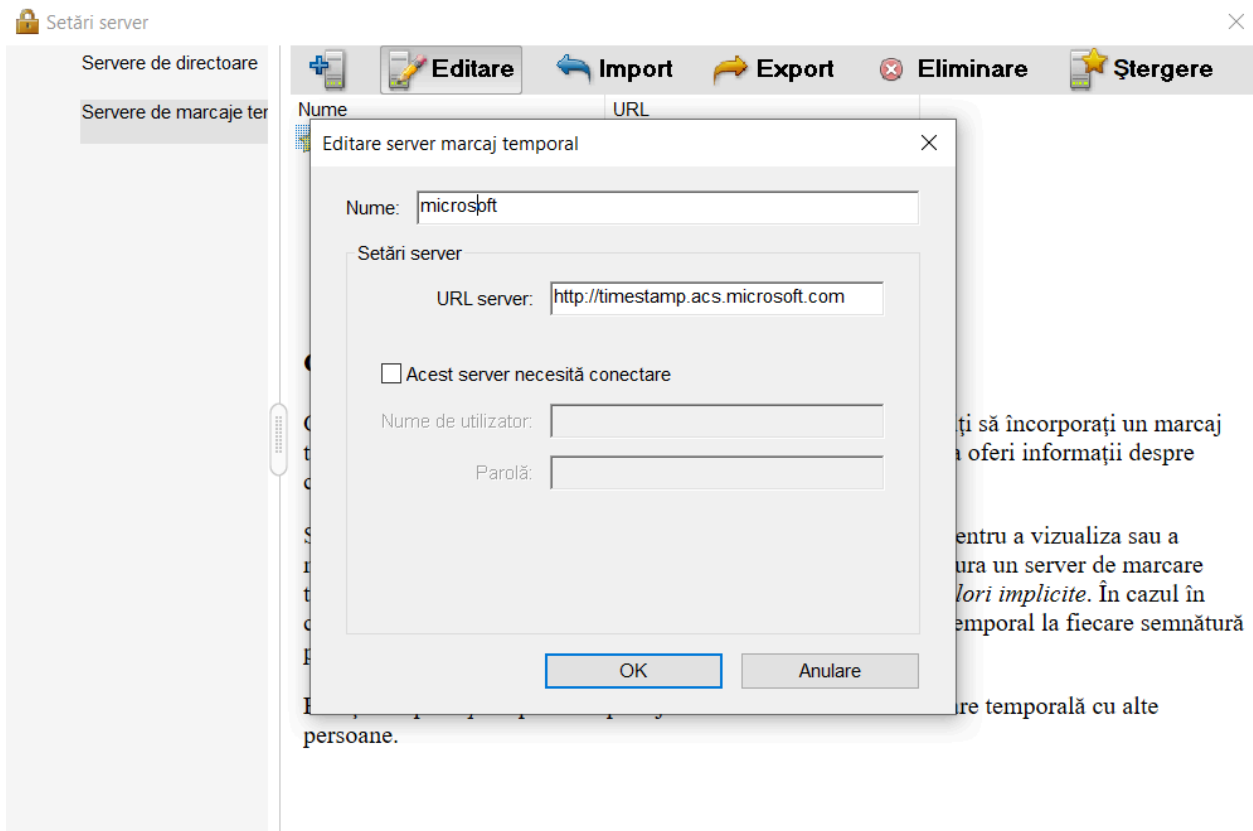
Continuare



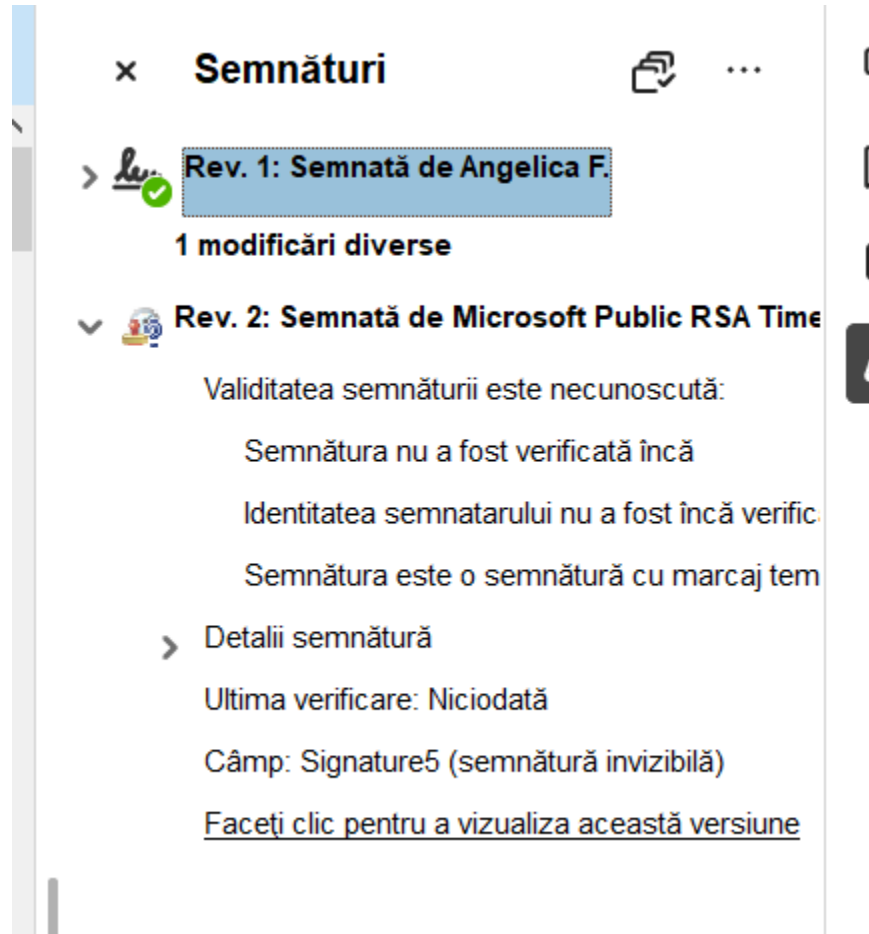
3.2. Validate a document



- 3.3. Search for a free timestamp server. Hint: github
- 3.4. Apply a timestamp to the document

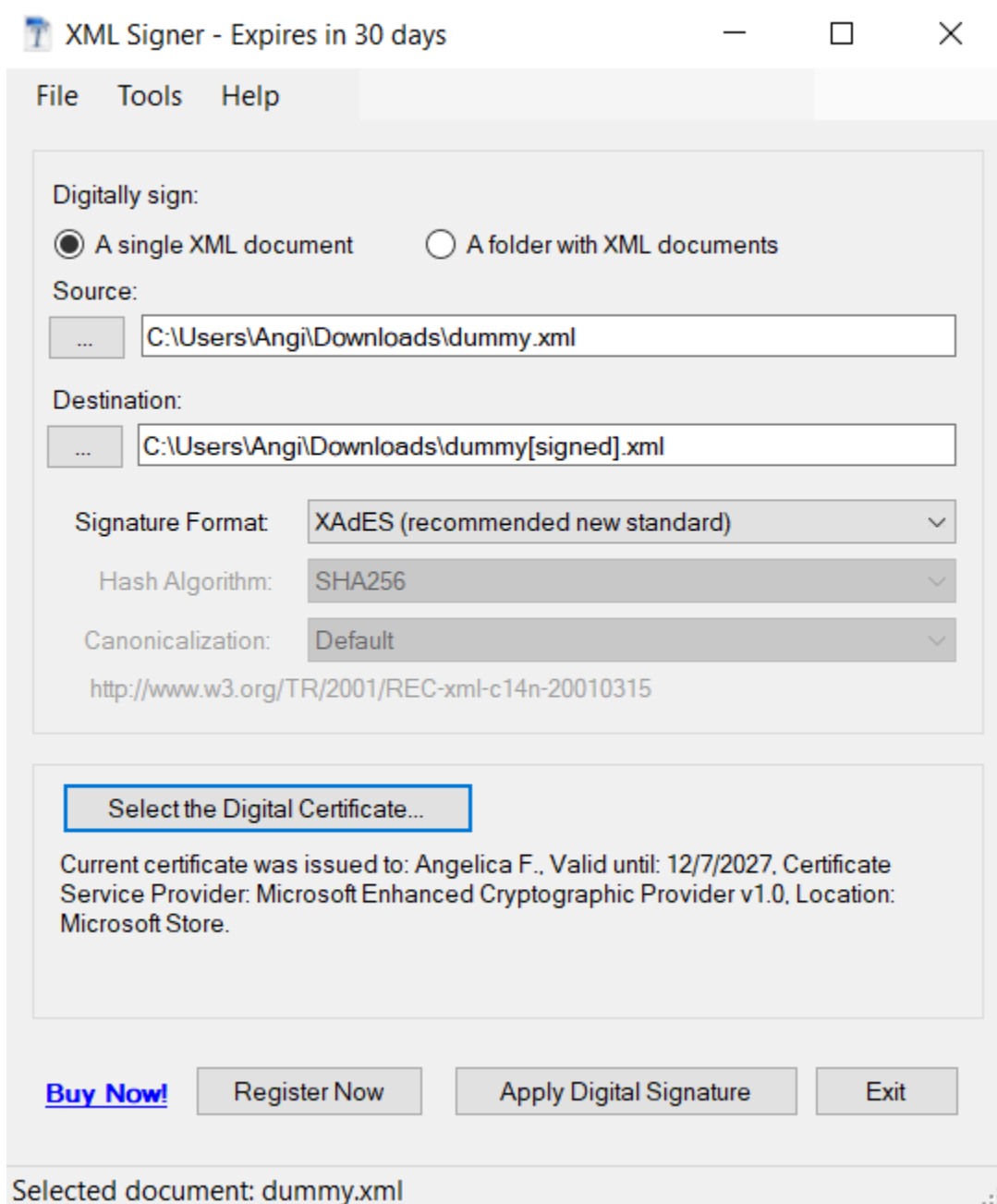


3.5. Validate the document and check the signature format this time

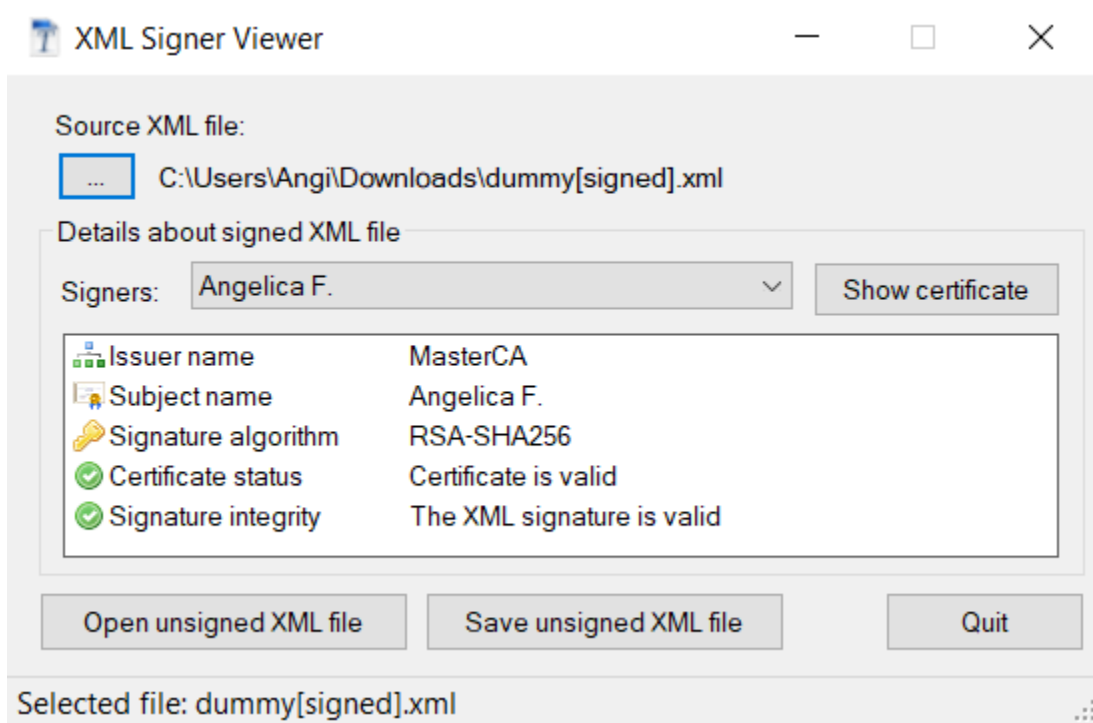


4. XML Signatures

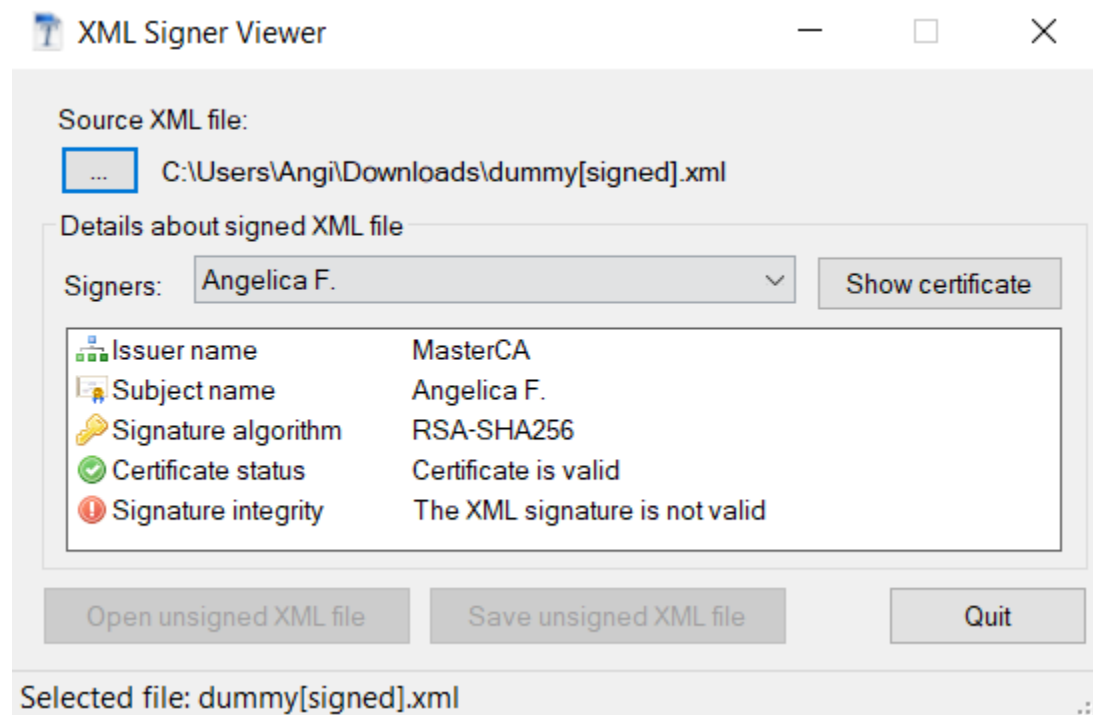
4.1. Sign an XML document



4.2. Validate the signed document



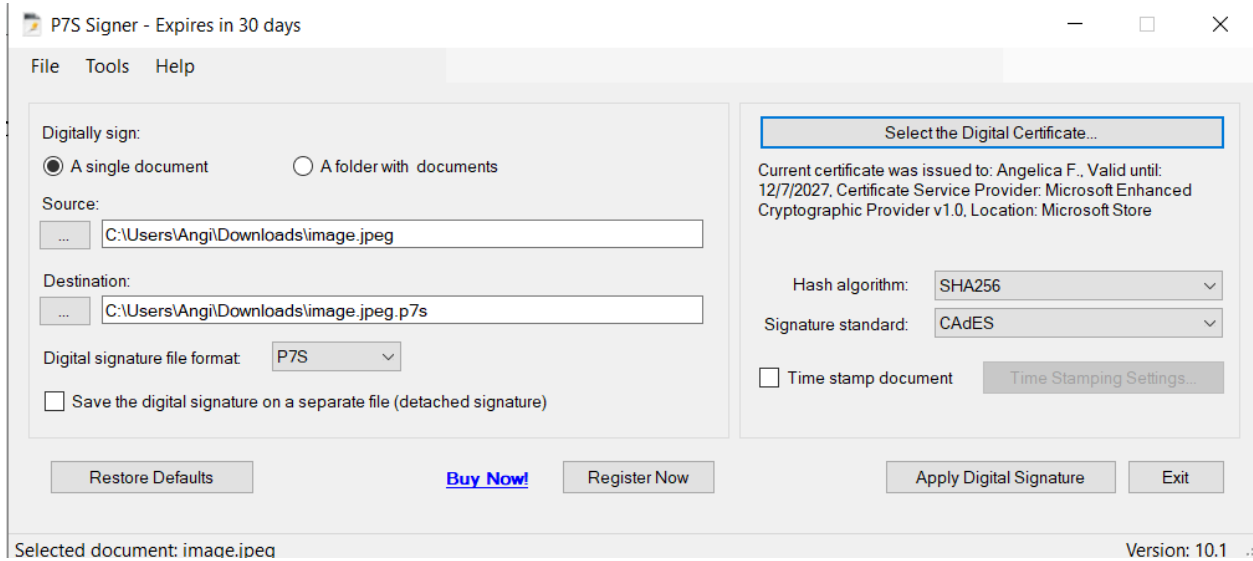
4.3. Validate an altered version of the document



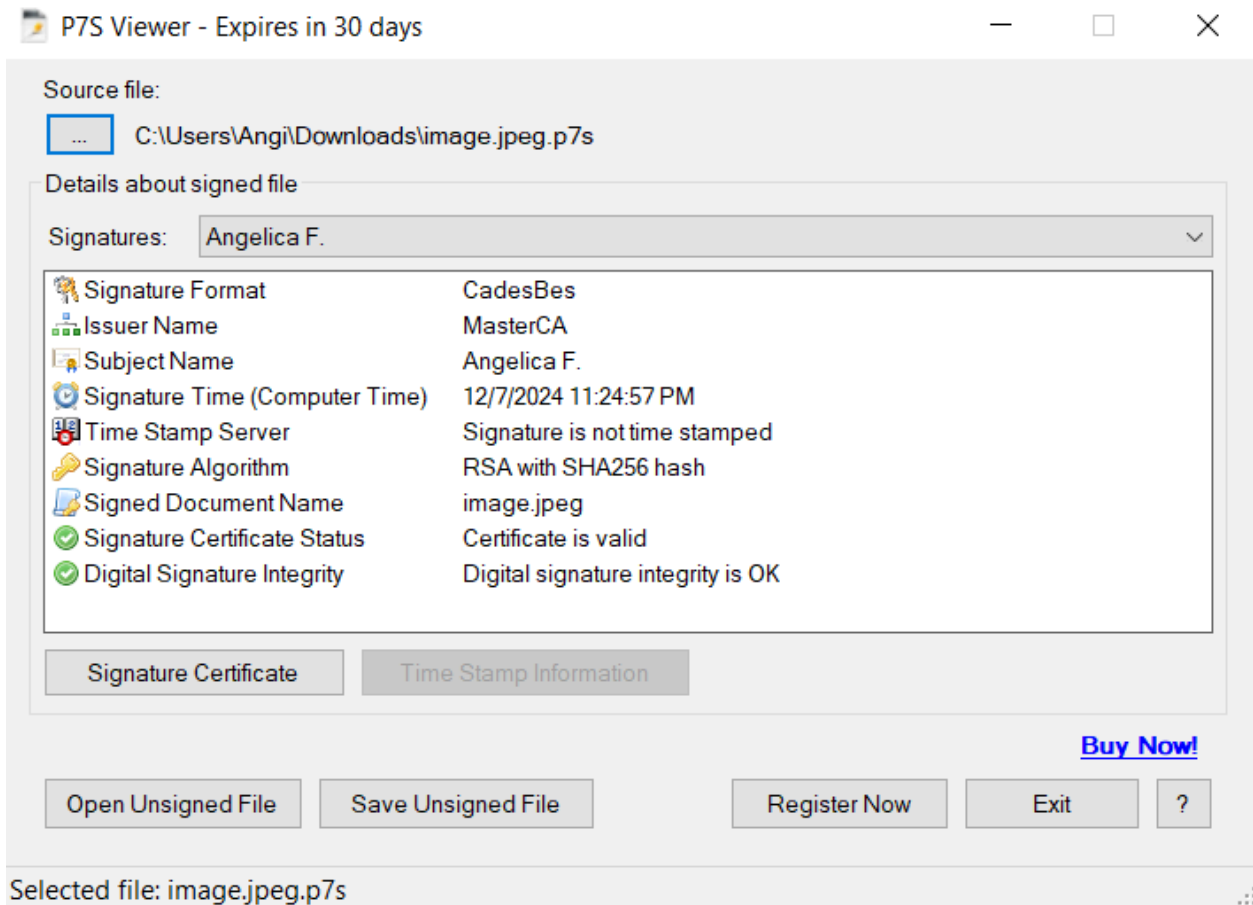
5. PKCS#7 Signatures

5.1. Use a media document - a picture, a text file, or other file type.

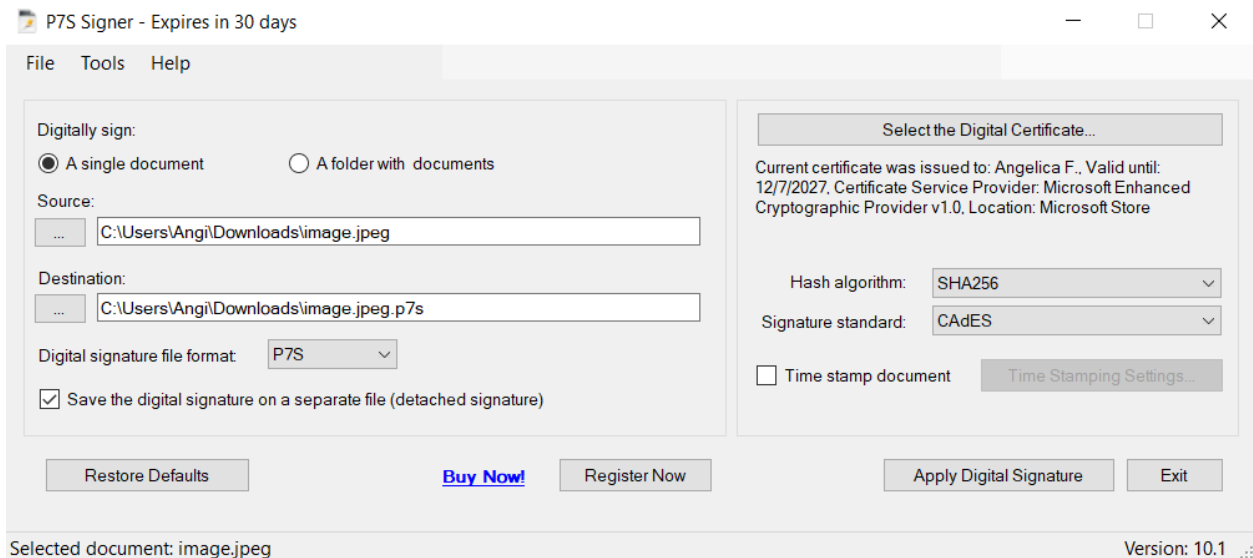
5.2. Obtain an attached signature



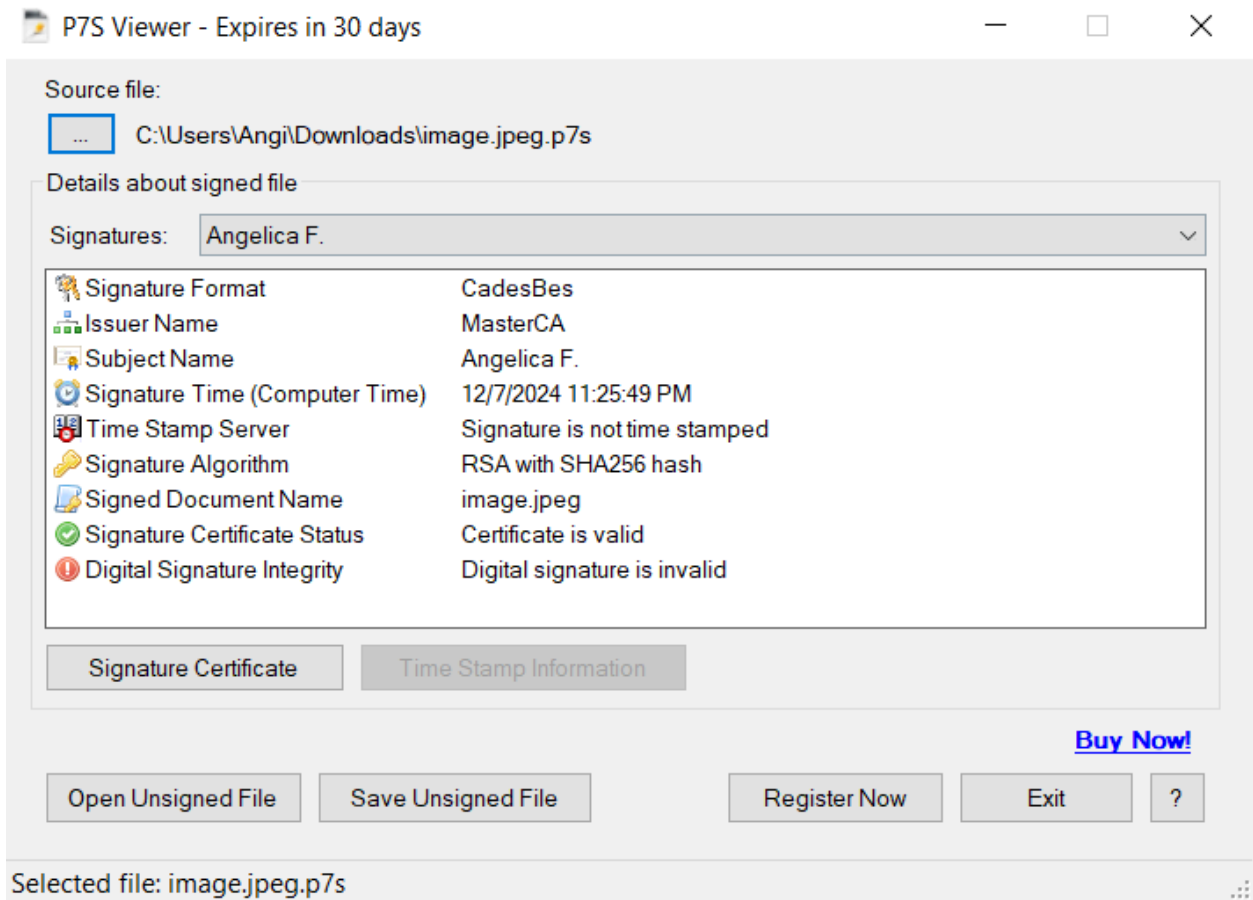
5.3. Validate it



5.4. Obtain a detached signature



5.5. Validate it



5.6. Check the obtained file with ASN.1 viewer.

```
C:\Users\Angi\Downloads>openssl asn1parse -inform DER -in image.jpeg.p7s
 0:d=0 hl=2 l=inf cons: SEQUENCE
 2:d=1 hl=2 l= 9 prim: OBJECT                :pkcs7-signedData
13:d=1 hl=2 l=inf cons: cont [ 0 ]
15:d=2 hl=2 l=inf cons: SEQUENCE
17:d=3 hl=2 l= 1 prim: INTEGER                :01
20:d=3 hl=2 l= 15 cons: SET
22:d=4 hl=2 l= 13 cons: SEQUENCE
24:d=5 hl=2 l= 9 prim: OBJECT                :sha256
35:d=5 hl=2 l= 0 prim: NULL
37:d=3 hl=2 l=inf cons: SEQUENCE
39:d=4 hl=2 l= 9 prim: OBJECT                :pkcs7-data
50:d=4 hl=2 l= 0 prim: EOC
52:d=3 hl=2 l=inf cons: cont [ 0 ]
54:d=4 hl=4 l= 925 cons: SEQUENCE
58:d=5 hl=4 l= 645 cons: SEQUENCE
62:d=6 hl=2 l= 3 cons: cont [ 0 ]
64:d=7 hl=2 l= 1 prim: INTEGER                :02
67:d=6 hl=2 l= 20 prim: INTEGER              :6EE1DC06A45F402A41ECDEA996B9601FC664FDDF
89:d=6 hl=2 l= 13 cons: SEQUENCE
91:d=7 hl=2 l= 9 prim: OBJECT                :sha256WithRSAEncryption
102:d=7 hl=2 l= 0 prim: NULL
104:d=6 hl=2 l= 19 cons: SEQUENCE
106:d=7 hl=2 l= 17 cons: SET
108:d=8 hl=2 l= 15 cons: SEQUENCE
110:d=9 hl=2 l= 3 prim: OBJECT                :commonName
115:d=9 hl=2 l= 8 prim: UTF8STRING            :MasterCA
125:d=6 hl=2 l= 30 cons: SEQUENCE
127:d=7 hl=2 l= 13 prim: UTCTIME              :241207192635Z
142:d=7 hl=2 l= 13 prim: UTCTIME              :271207192634Z
157:d=6 hl=2 l= 81 cons: SEQUENCE
159:d=7 hl=2 l= 20 cons: SET
161:d=8 hl=2 l= 18 cons: SEQUENCE
163:d=9 hl=2 l= 3 prim: OBJECT                :commonName
168:d=9 hl=2 l= 11 prim: UTF8STRING            :Angelica F.
181:d=7 hl=2 l= 17 cons: SET
183:d=8 hl=2 l= 15 cons: SEQUENCE
185:d=9 hl=2 l= 3 prim: OBJECT                :givenName
190:d=9 hl=2 l= 8 prim: UTF8STRING            :Angelica
200:d=7 hl=2 l= 11 cons: SET
202:d=8 hl=2 l= 9 cons: SEQUENCE
204:d=9 hl=2 l= 3 prim: OBJECT                :surname
209:d=9 hl=2 l= 2 prim: UTF8STRING            :F.
213:d=7 hl=2 l= 12 cons: SET
215:d=8 hl=2 l= 10 cons: SEQUENCE
217:d=9 hl=2 l= 3 prim: OBJECT                :organizationName
222:d=9 hl=2 l= 3 prim: UTF8STRING            :ASW
227:d=7 hl=2 l= 11 cons: SET
229:d=8 hl=2 l= 9 cons: SEQUENCE
231:d=9 hl=2 l= 3 prim: OBJECT                :countryName
```

6. Microsoft Word Signatures

6.1. Sign a document using Microsoft Word.

Signature Setup ? X

Suggested signer (for example, John Doe):
Angelica Flutur

Suggested signer's title (for example, Manager):
Student

Suggested signer's e-mail address:
fluturangelica24@stud.ase.ro

Instructions to the signer:
Before signing this document, verify that the content you are signing is correct.

☐ Allow the signer to add comments in the Sign dialog

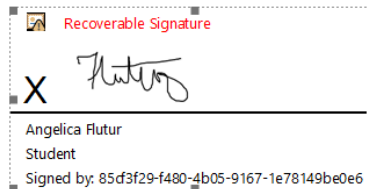
☒ Show sign date in signature line

OK Cancel

ent as final to discourage editing.

Edit Anyway

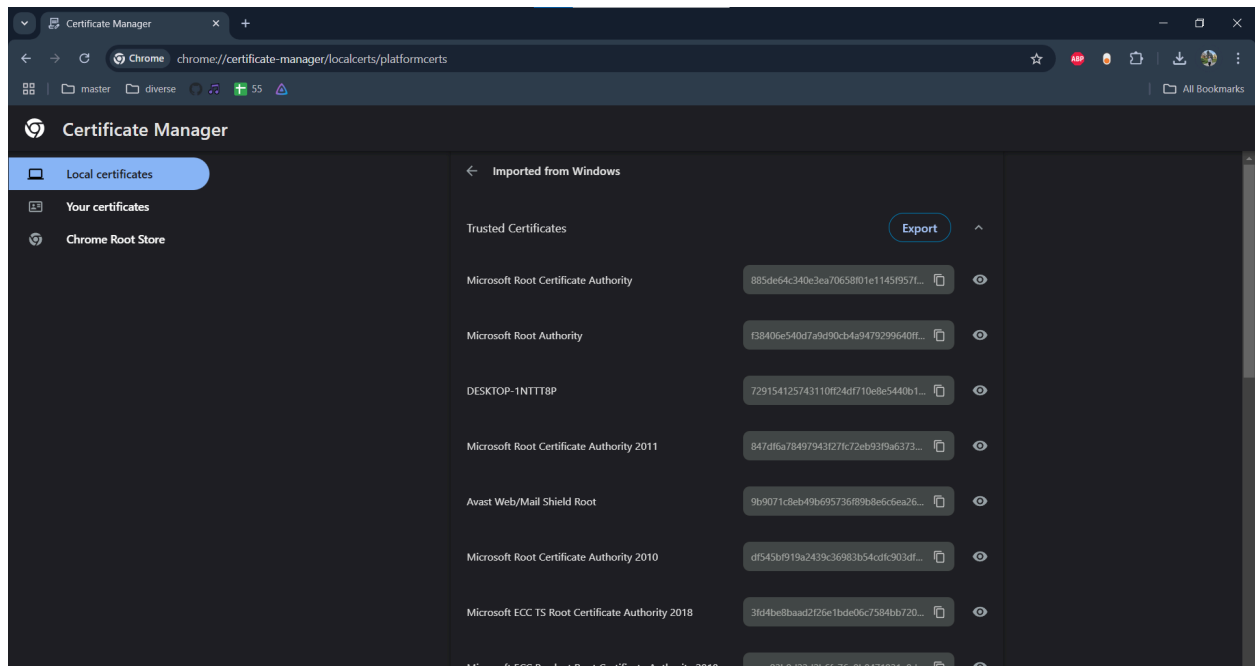
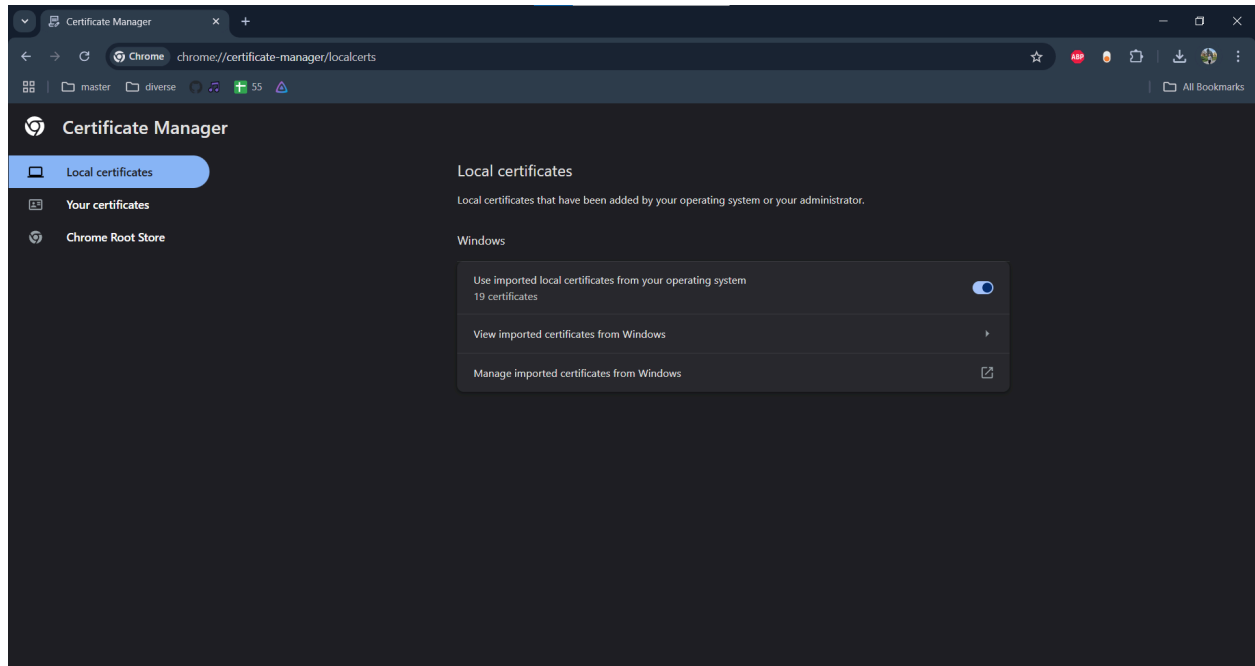
Sign a document using Microsoft Word.

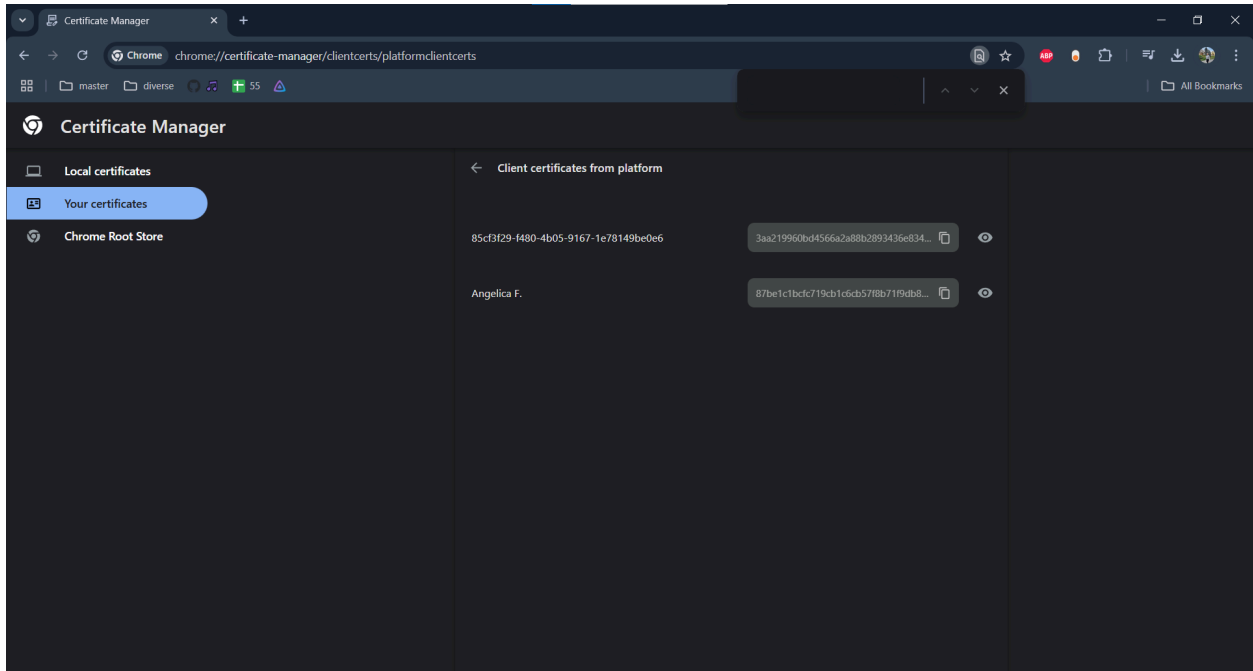


7. Mozilla vs Microsoft Store

- 7.1. Prove that we are using different stores to validate the certificate of the web server.
- 7.2. We can use <https://testssl.certsign.ro/>
- 7.3. Hint: Should use different browsers (Mozilla+Chrome, for example)

CHROME:





Certificate Viewer: Angelica F.

General

Details

Issued To

Common Name (CN)

Organization (O)

Organizational Unit (OU)

Angelica F.

ASW

<Not Part Of Certificate>

Issued By

Common Name (CN)

Organization (O)

Organizational Unit (OU)

MasterCA

<Not Part Of Certificate>

<Not Part Of Certificate>

Validity Period

Issued On

Expires On

Saturday, December 7, 2024 at 9:26:35 PM

Tuesday, December 7, 2027 at 9:26:34 PM

SHA-256 Fingerprints

Certificate

Public Key

87be1c1bcfc719cb1c6cb57f8b71f9db8302906ad9c66308c351ea5fb18abf2c

6696110f807da3469630f11e1ebd5f54c31ac8d7cc7bacf89d664158310620fe

FIREFOX:

Firefox

about:preferences#searchResults

🔍

Your browser is being managed by your organization.

cert

Search Results

Certificates

☒ Query OCSP responses for certificates

Certificate Manager

×

Your Certificates

Authentication Decisions

People

Servers

Authorities

You have certificates on file that identify these certificate authorities

Certificate Name	Security Device	
> ACCV		⬆
> Actalis S.p.A./03358520967		
> AffirmTrust		
> Agence Nationale de Certification Electr...		
> Amazon		
> ANF Autoridad de Certificacion		⬇

View...

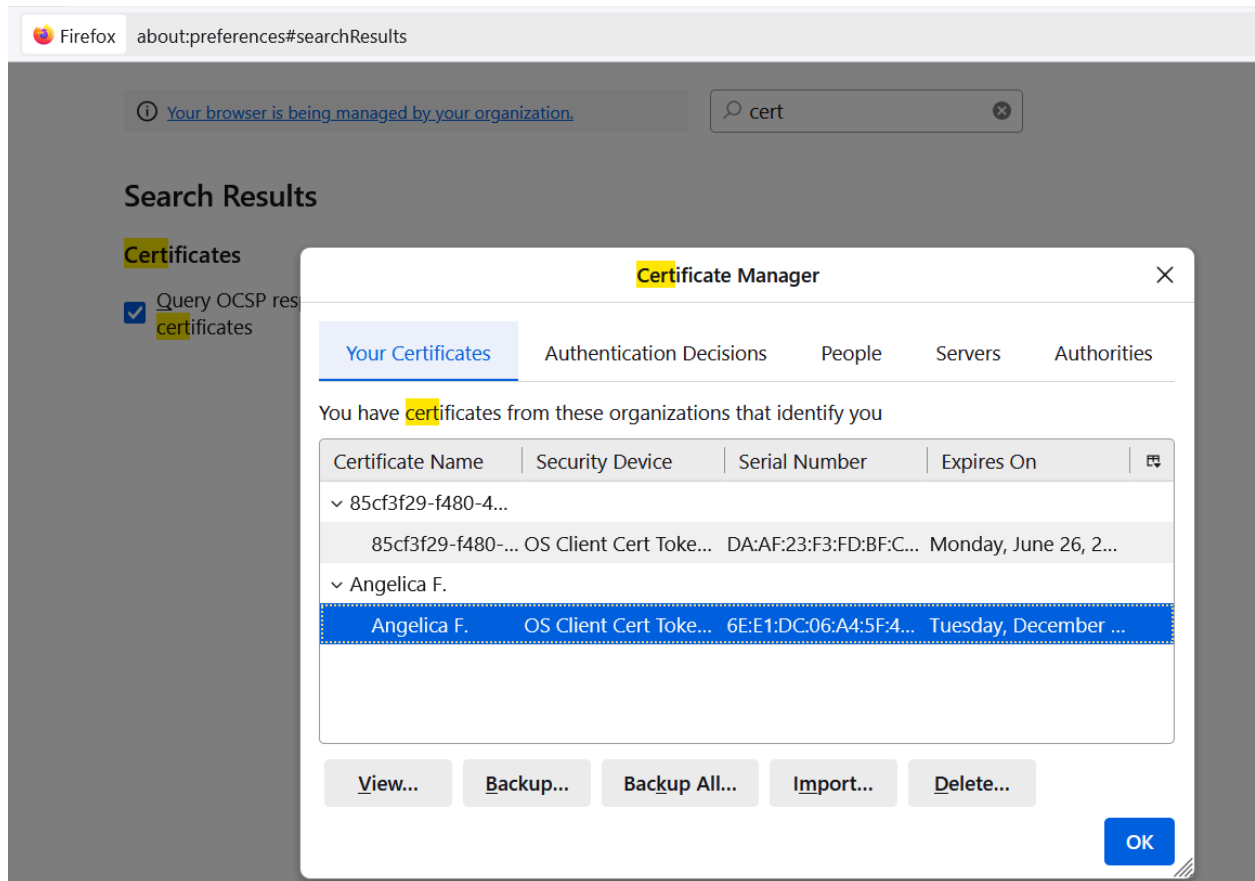
Edit Trust...

Import...

Export...

Delete or Distrust...

OK



CONCLUSION:

The difference between the 2 stores is that Chrome is based on the Microsoft Certificate Store, while Firefox has its own certificate store.

8. Certificate Requests

8.1. Generate a certificate request using openssl.

[illegible]

8.2. Use the generated CSR to issue a new certificate, from EJBCA

EJBCA Administration x +

← → ↺ 🏠 <https://localhost:8443/ejbcadminweb/ra/addendentity.jsp>

EJBCA Kali Webgoat TLS Page

 **EJBCA**
PKI by PrimeKey

Home

CA Functions

- CA Activation
- CA Structure & CRLs
- Certificate Profiles
- Certification Authorities
- Crypto Tokens
- Publishers
- Validators

RA Functions

- Add End Entity
- End Entity Profiles
- Search End Entities
- User Data Sources

Supervision Functions

- Approval Profiles
- Approve Actions

System Functions

- Roles and Access Rules
- Internal Key Bindings
- Services

System Configuration

- CMP Configuration
- EST Configuration
- SCEP Configuration
- System Configuration

My Preferences

RA Web

Public Web

Logout

Add End Entity

End Entity Profile	User	Required
Username	angelicaflutur	☒
Password (or Enrollment Code)	••••••	☒
Confirm Password	••••••	
E-mail address	flutur.angelica@gmail.com	☐
Subject DN Attributes		
CN, Common name	ANGELICA FLUTUR	☒
givenName, Given name (first name)	ANGELICA	☐
surname, Surname (last name)	FLUTUR	☐
C, Country (ISO 3166)	RO	☒
O, Organization	ASE	☐
Main Certificate Data		
Certificate Profile	TLS_Client	☒
CA	IssuingCA	☒
Token	User Generated	☒
Add Reset		

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Enroll

- Create Certificate from CSR
- Create Keystore
- Create CV certificate

Register

- Request Registration

Retrieve

- Fetch CA Certificates
- Fetch CA CRLs
- List User's Certificates
- Fetch User's Latest Certificate

Inspect

- Inspect certificate/CSR
- Check Certificate Status

Miscellaneous

- Administration

The EJBCA Public Web has been deprecated and will be removed in an upcoming version of EJBCA. Please move your workflows to the EJBCA RA UI

Certificate enrollment from a CSR

Please give your username and enrollment code, select a PEM- or DER-formatted certification request file (CSR) for upload, or paste a PEM-formatted

A PEM-formatted request is a BASE64 encoded certificate request starting with

-----BEGIN CERTIFICATE REQUEST-----
and ending with
-----END CERTIFICATE REQUEST-----

If the CA signing the request is an External CA, choose "PEM - certificate only" as the result type.

Enroll

Username:

Enrollment code:

Request file: openssl.csr

or pasted request

Result type:

The EJBCA Public Web has been deprecated and will be removed in an upcoming version of EJBCA. Please move your workflows to the EJBCA RA UI

Certificate Created

Subject DN: CN=ANGELICA FLUTUR,GIVENNAME=ANGELICA,SURNAME=FLUTUR,O=ASE,C=RO

Issuer DN: CN=MasterCA

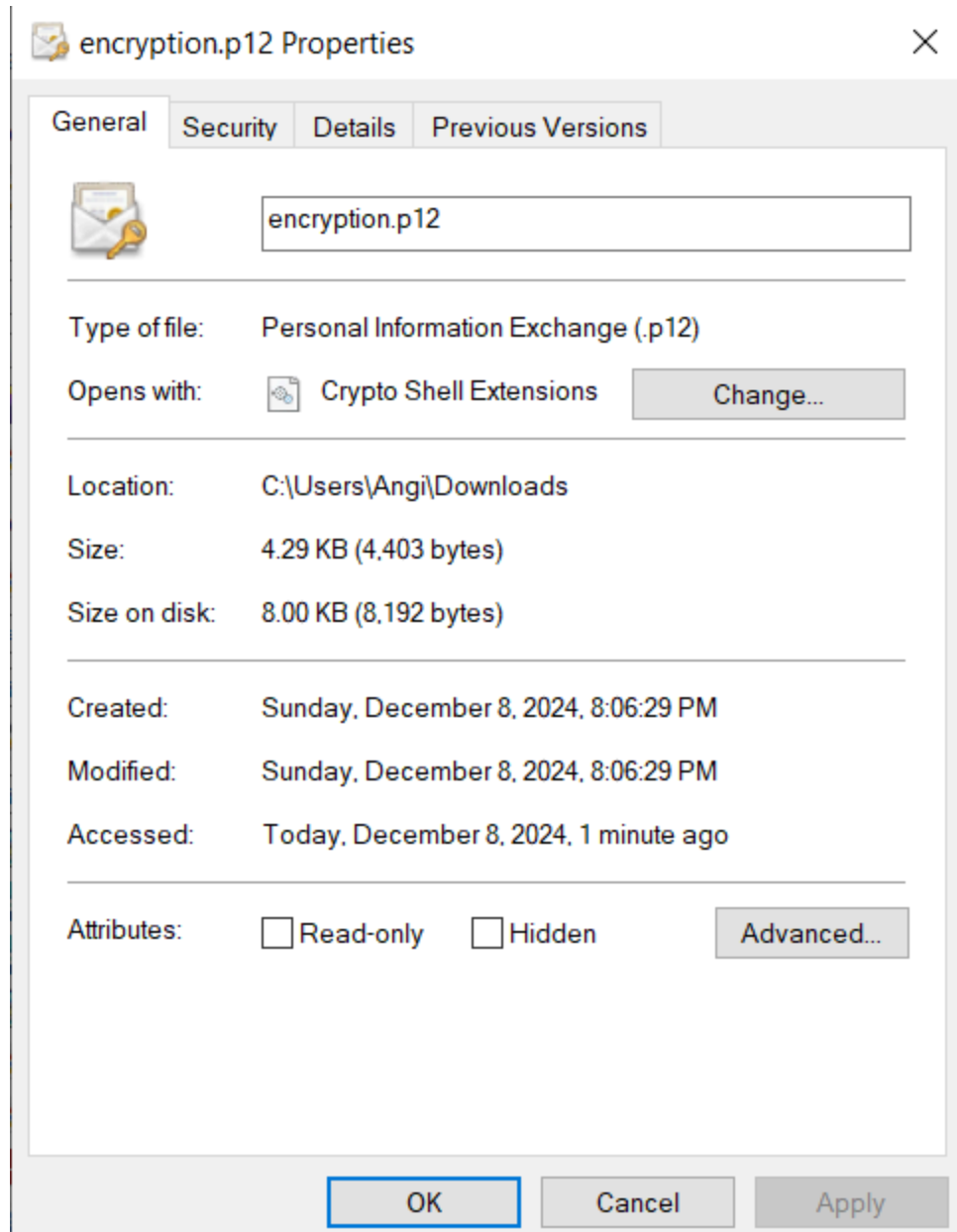
Serial Number: 518B5DCBBA53E330749DC57A5F02342F276D323

You should receive your certificate file in a few seconds. If nothing happens, click this link:

[Download certificate](#)

8.3. Use openssl to create a pkcs12 file from certificate and private key

```
C:\Users\Angi\Downloads>openssl pkcs12 -export -in ANGELICAFLUTUR.pem -inkey private.key -out encryption.p12
Enter pass phrase for private.key:
Enter Export Password:
Verifying - Enter Export Password:
```



8.4. Import the certificate to Windows store

8.5. Try to sign a PDF file Useful commands:

- openssl req -newkey rsa:2048 -keyout private.key -out openssl.csr
- openssl pkcs12 -export -in encryption.crt -inkey enc_private.key -out encryption.p12

Semnați cu un ID digital



Alegeți ID-ul digital pe care doriți să îl utilizați pentru semnare:

Reîmprospătare



Angelica F. (ID digital Windows)
Emis de: MasterCA, Expiră: 2027.12.07

[Vizualizare detalii](#)



ANGELICA FLUTUR (ID digital Windows)
Emis de: MasterCA, Expiră: 2027.12.08

[Vizualizare detalii](#)



Configurare ID digital nou

Anulare

Continuare

Electronic_Signatures... x

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Semnat, cu toate semnăturile valide.

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- Scanare & OCR
- Compresie PDF
- Completare și semnare
- Adăugare ștampilă
- Utilizare certificat
- Utilizare producere imprimare
- Măsurare obiecte
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Bonus

Electronic Signatures - Tasks

Julian Aciobanitei, Phd

Bucharest
2024

ANGELICA FLUTUR

Semnat digital de ANGELICA FLUTUR
Data: 2024.12.08
20:09:07
+02'00'

Semnături

Rev. 1: Semnată de ANGELICA FLUTUR

Semnătura este validă:
Documentul nu a fost modificat după aplicare
Semnat de utilizatorul curent
Semnătura include un marcat temporal încorporat
Semnătura este activată pentru LTV

Detalii semnătură

Detalii certificat...
Ultima verificare: 2024.12.08 20:10:12 +02'00'
Câmp: Signature2 la pagina 1
[Faceți clic pentru a vizualiza această versiune](#)

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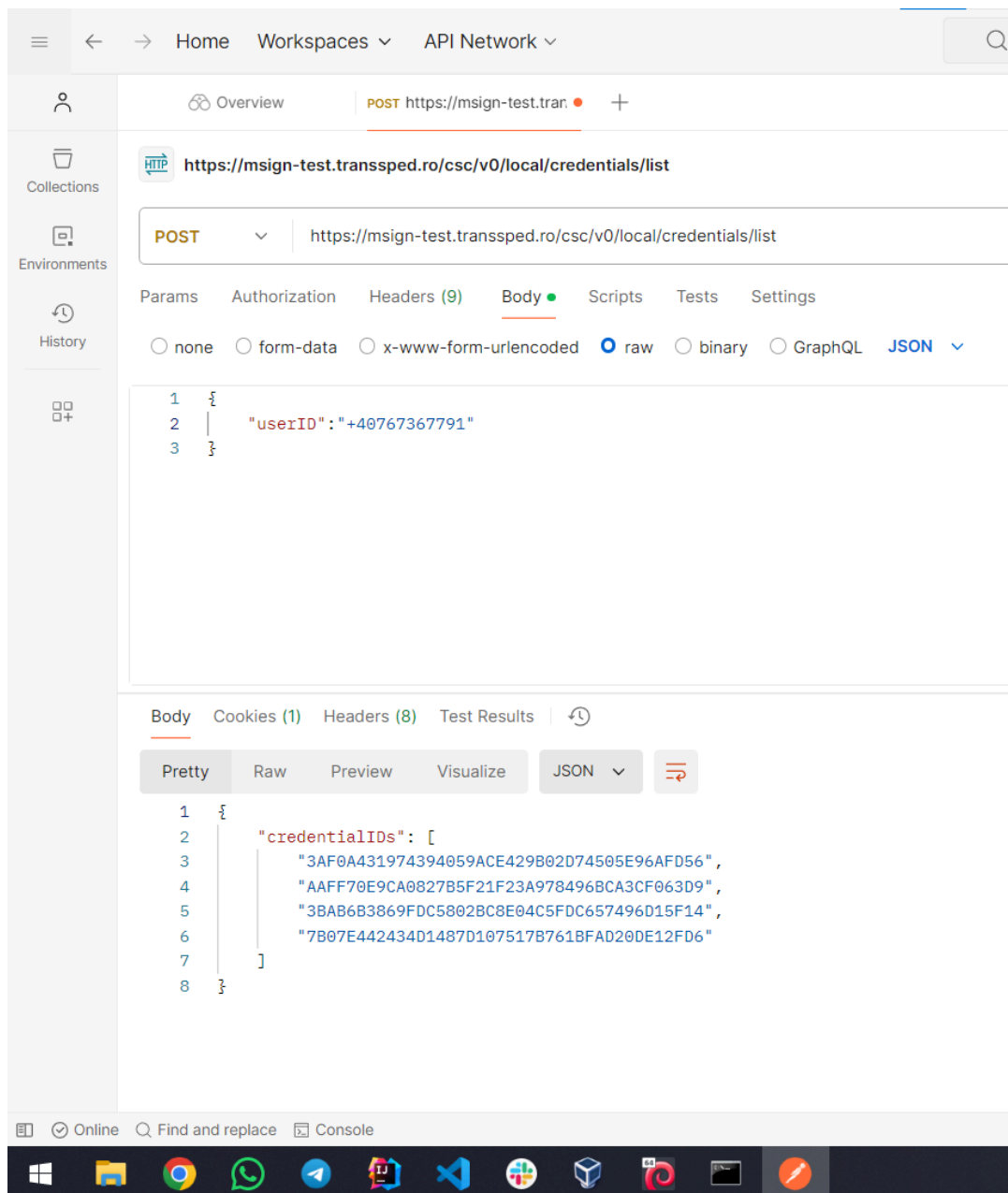
8:10 PM
12/8/2024

9. CSC Protocol - 'Hello world' level

Obtain one of the cloud certificates of the user with phone number +40767367791, using the CSC protocol.
Helping details:

- 9.1. CSC API is exposed here (<https://msign-test.transsped.ro/csc/v0/local/>).
- 9.2. CSC API documentation can be found here (<https://drive.google.com/file/d/1S1MZ3NNhfhLYlXgVcdFxE19OS4h6KHrW/view>).
- 9.3. You need to call /credentials/list and credentials/info

<https://msign-test.transsped.ro/csc/v0/local/credentials/list>



<https://msign-test.transped.ro/csc/v0/local/credentials/info>

[illegible]